diff-gpt: autoregressive time-series forecasting via derivative tokenization with MASH $\Sigma\Delta$-quantization

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Abstract

We study autoregressive forecasting of continuous time series with a decoder-only transformer trained on a derivative-based tokenization. The encoder differentiates the input to order $k \geq 1$, scales by its observed range, and quantizes with a *MASH* (multi-stage noise-shaping) k-th-order $\Sigma\Delta$ modulator; the decoder inverts this. We prove two tokenization-level bounds, both independent of the model: the achievable cross-entropy loss grows linearly in $\log V$, and the reconstruction error is uniformly bounded by $\Delta/2 = D/(V - 2 - 2^k)$ for any k such that $V > 2^k + 2$. We further observe that while MASH's reconstruction bound is tight against pure quantization noise, *model prediction error* compounds as $O(\Delta \cdot T^{k-1})$ through k decoder integrations — giving a principled reason to prefer $k = 1$ for forecasting despite the clean higher-order theory. We introduce an ordinal Gaussian soft cross-entropy that converts bin-wise classification into a distance-aware loss over ordinal bins; on M4 Hourly it reduces sMAPE from 18.58 to 17.49 (6% relative) without architectural change.

1 Introduction

Most state-of-the-art time-series forecasters are regression models with specialized inductive biases (patching, inverted attention, channel mixing) trained against an ℓ_2 or Huber loss. By contrast, large language models are decoder-only transformers trained with cross-entropy over a discrete vocabulary. The question we address is whether a pure language-model recipe — fixed architecture, fixed loss, autoregressive generation — is competitive on standard forecasting benchmarks, given a faithful discrete representation of the continuous input.

The key choice is the tokenization. We describe a derivative-based encoder with MASH $\Sigma\Delta$ quantization that satisfies two closed- form vocabulary-scaling laws: (a) the ideal cross-entropy loss grows linearly in $\log V$ (Prop. 7) and (b) the reconstruction error between a round-trip of encoder and decoder is bounded uniformly in T by $\Delta/2$ for any order k (Thm. 1). Both are tokenization-level statements — they hold regardless of what model sits on top.

The rest of the paper. Section 2 describes the encoder and decoder; Section 3 proves the two scaling laws and warns that the reconstruction theorem's tight bound is against quantization noise only — the model's prediction error compounds less kindly at higher k; Section 4 introduces ordinal soft cross-entropy and motivates it from the differential-entropy bound; Section 5 reports M4 Hourly and ETTh1 results.

2 Derivative tokenization

Let $x \in \mathbb{R}^{T \times F}$ be a multivariate signal with F channels of length T. We fix a derivative order $k \geq 1$ and define $D_c = \max_t |x_{t+1,c} - x_{t,c}|$ per channel (the *domain of definition*). For even V with $V > 2^k + 2$, define the per-channel scale

$$s_c = \frac{V - 2 - 2^k}{2D_c}, \qquad \Delta_c = \frac{1}{s_c} = \frac{2D_c}{V - 2 - 2^k}. \tag{1}$$

The -2^k headroom is the maximum overshoot MASH can produce over the unshaped input (Lemma 2); without it the quantizer would saturate at signal peaks and lose the noise-shaping guarantee.

2.1 MASH k-th-order modulator.

Each of the k stages is a stable first-order $\Sigma\Delta$ loop. Stage 1 operates on the scaled k-th finite difference $y_t = s \cdot \Delta^k x_t$; stage $i \geq 2$ operates on $-\varepsilon_t^{(i-1)}$, the previous stage's quantization error (bounded by $\frac{1}{2}$ in bin units). Let $q_t^{(i)}$ denote stage i's integer output and $\varepsilon_t^{(i)} = q_t^{(i)} -$ (stage i's quantizer input).

The final token stream is the digital-differentiator combination

$$q_t \;=\; \sum_{i=0}^{k-1} (1 - z^{-1})^i \, q_t^{(i+1)} \;=\; q_t^{(1)} + \Delta q_t^{(2)} + \Delta^2 q_t^{(3)} + \cdots + \Delta^{k-1} q_t^{(k)}, \tag{2}$$

shifted by $V/2$ and clipped to $\{0, \ldots, V-1\}$.

2.2 Decoder.

Given the start prefix $(x_1, \ldots, x_k)$ and token sequence (q_t), the decoder reconstructs the signal by dequantizing and integrating k times:

$$\hat{x}_{t+1} \;=\; \hat{x}_t + (q_t - V/2)\,\Delta. \tag{3}$$

2.3 Per-column orders.

For multivariate input where channels have qualitatively different dynamics (e.g., a slowly-drifting trend column alongside a stationary-noise column), a single shared k is suboptimal. The implementation accepts k as a per-column vector k_c instead of a scalar. Each column is differentiated to its own k_c, scaled by $(V - 2 - 2^{k_c})/(2D_c)$ if $k_c \geq 2$ (for MASH headroom) or $(V-2)/(2D_c)$ otherwise, then passed through the appropriate $\Sigma\Delta$ variant per column (plain round for $k_c = 0$, carry scheme for $k_c = 1$, MASH for $k_c \geq 2$). The encoder aligns all columns to the shared valid window $T - \max_c k_c$, stores the raw prefix as *start*, and the decoder reconstructs each column using only its own k_c cumsums. Theorem 1 and Prop. 7 apply per column with D_c, k_c replacing their scalar counterparts.

3 Tokenization contract

Two tokenization-level results, both independent of the model.

3.1 Reconstruction bound

Theorem 1 (MASH reconstruction, any k)**.** *For any signal with $D = \max_t |\Delta^k x_t| > 0$ and vocabulary size $V > 2^k + 2$, the MASH encoder–decoder pair satisfies*

$$\|\hat{x} - x\|_\infty \;\leq\; \frac{D}{V - 2 - 2^k} \;=\; \frac{\Delta}{2}, \qquad \textit{uniformly in } T. \tag{4}$$

The proof relies on a standard identity about the cascade.

Lemma 2 (MASH noise-shaping)**.** *Let $\varepsilon_k := \varepsilon^{(k)}$ be the quantization error of the final stage. The combined output* (2) *satisfies*

$$q_t - y_t \;=\; (1 - z^{-1})^k \, \varepsilon_{k,t}, \qquad |\varepsilon_{k,t}| \;\leq\; \tfrac{1}{2}. \tag{5}$$

Sketch. Induction on the cascade. Stage 1's first-order $\Sigma\Delta$ gives $q_t^{(1)} = y_t + \Delta\varepsilon_t^{(1)}$. Stage $i \geq 2$ with input $-\varepsilon_t^{(i-1)}$ gives $q_t^{(i)} = -\varepsilon_t^{(i-1)} + \Delta\varepsilon_t^{(i)}$. Substituting into (2) and telescoping the intermediate $\varepsilon^{(i)}$ terms yields $q_t = y_t + \Delta^k \varepsilon_{k,t}$. The bound $|\varepsilon_{k,t}| \leq 1/2$ is immediate: each 1st-order stage rounds to the nearest integer, so its internal quantization error is in $(-\frac{1}{2}, \frac{1}{2}]$. □

Lemma 3 (bounded MASH output)**.** $|q_t - y_t| \leq 2^{k-1}$.

Proof. $|\Delta^k \varepsilon_k| \leq \|(1 - z^{-1})^k\|_1 \cdot \max|\varepsilon_k| = 2^k \cdot \frac{1}{2} = 2^{k-1}$. □

The -2^k headroom in the scale's denominator ensures this overshoot stays within the quantizer range, avoiding saturation.

Proof of Theorem 1. The decoded increment is $(q_t - V/2)\Delta = (y_t + \Delta^k \varepsilon_{k,t})\Delta$. The true increment is $y_t\Delta$. Their difference is $\Delta^k \varepsilon_{k,t} \cdot \Delta$. Summing k times from the exact starts telescopes the Δ^k down to $\varepsilon_{k,t}$:

$$\hat{x}_T - x_T \;=\; \varepsilon_{k,T-k} \cdot \Delta. \tag{6}$$

By Lemma 2, $|\varepsilon_k| \leq \frac{1}{2}$, so $|\hat{x}_T - x_T| \leq \Delta/2$. □

Corollary 4 (scaling)**.** $\|\hat{x} - x\|_\infty = O(D/V)$. *Vocabulary doubling halves the reconstruction error, uniformly in T and for any k.*

3.2 Prediction-error compounding vs. quantization noise

Theorem 1 bounds the residual between $\hat{x}$ and x *given* a perfect encoder/decoder round-trip. In autoregressive forecasting, the decoder does not consume the exact encoded tokens of the true signal; it consumes tokens *predicted* by the model, which carry prediction error on top of quantization noise.

Proposition 5 (prediction-error amplification at order k)**.** *Suppose the model's per-step prediction error has typical magnitude η bins (so $\eta \cdot \Delta$ in value space per step). After k decoder integrations the forecast error at the horizon end grows as*

$$|\hat{x}_{T+H} - x_{T+H}| \;=\; O\Big(\eta \cdot \Delta \cdot H^{\,k-1}\Big), \tag{7}$$

where H is the number of forecast steps.

Sketch. Token prediction error δq_t of typical size η enters the decoder as diff-estimate error $\delta q_t \cdot \Delta$. Under k successive cumsum operations, an i.i.d. or coherent bias sequence of size $\eta\Delta$ produces output whose magnitude at step H scales as $\eta\Delta H^{k-1}$ (integration of polynomial in H). □

Remark 6 ($k = 1$ preferred for forecasting)*.* At $k = 1$ the amplification is $H^0 = 1$: prediction errors contribute proportionally to their bin size, not multiplicatively across the horizon. At $k = 2$ a single-bin token error grows $H \cdot \Delta$ over a 48-step forecast, on the order of 24Δ — much larger than the $\Delta/2$ quantization bound. Empirically (Section 5) this manifests as catastrophic forecasting regression at $k = 2$ despite strictly lower token-level cross-entropy.

3.3 Cross-entropy bound

Let X_t be the continuous next-step increment conditional on context ctx, with density $p(\cdot \mid \text{ctx})$ and differential entropy $h(X_t \mid \text{ctx}) = -\int p \log p$.

Proposition 7 (achievable CE)**.** *In the fine-quantization limit $\Delta \to 0$, the expected cross-entropy of the optimal autoregressive token predictor satisfies*

$$\mathcal{L}^\star(V) \;=\; h(X_t \mid \mathrm{ctx}) + \log V - \log(2D) \;+\; o(1). \tag{8}$$

Equivalently, $\mathcal{L}^\star$ is linear in $\log V$ with slope 1.

Sketch. Standard quantized-entropy relation (Cover & Thomas, §8.3): for a continuous RV with density p and bin width Δ,

$$H(\mathrm{bin}) \;=\; h(X) - \log \Delta + o(1), \qquad \Delta \to 0. \tag{9}$$

Condition on ctx and substitute $\Delta = 2D/(V - 2 - 2^k) \sim 2D/V$. □

Corollary 8 (vocabulary-invariant skill)**.** *Two models at vocabulary sizes $V_1 \neq V_2$ are comparable only through the* differential-entropy residual

$$\tilde{h}(V) \;:=\; \mathcal{L}(V) - \log V + \log(2D), \tag{10}$$

which estimates $h(X_t \mid \mathrm{ctx})$ and is independent of V at the rate.

3.4 Empirical check

M4 Hourly, global model, V-sweep with σ scaled $\propto V$ to hold value-space soft-target smoothing constant:

V	σ	val CE (nats)	$\Delta \log V$	Δ(CE)
64	0.5	1.01	—	—
128	1.0	1.70	$\log 2 = 0.69$	0.69
256	2.0	2.35	$\log 2 = 0.69$	0.65
512	4.0	3.04	$\log 2 = 0.69$	0.69

Empirical CE increments match $\log 2$ per vocab doubling to within 0.04 nat. Consistent with Prop. 7.

4 Ordinal soft cross-entropy

Standard cross-entropy treats the V bins as nominal categories: a prediction off by one bin is penalized identically to one off by fifty. Bins are an ordinal index over a continuous scale — off by one is almost right.

We replace the one-hot target $\mathbf{1}_{i=y}$ with a discrete Gaussian over bin indices:

$$p_i^{\mathrm{soft}}(y;\sigma) \;\propto\; \exp\!\left(-\tfrac{(i-y)^2}{2\sigma^2}\right), \qquad \sigma \geq 0, \tag{11}$$

and train with soft-target cross-entropy $\mathcal{L}_\sigma = -\sum_i p_i^{\mathrm{soft}} \log q_i$, where $q = \mathrm{softmax}(\mathrm{logits})$. Taking $\sigma \to 0$ recovers ordinary cross-entropy with one-hot targets.

Why this matters for a $\Sigma\Delta$-quantized signal. A continuous value landing near a bin boundary is, modulo the MASH carry, assigned to one of two adjacent bins; the encoder can shift either way depending on the running accumulator. One-hot training pretends this assignment is deterministic; soft targets acknowledge the effectively stochastic tie-breaking. More broadly, any aleatoric component of $p(X \mid \mathrm{ctx})$ makes the true conditional distribution over bins a smooth Gaussian-like shape; label smoothing with scale σ in bin units is the ordinal equivalent.

Bin-space versus value-space. Because bin width $\Delta \approx 2D/V$, a constant value-space smoothing $\sigma_{\text{val}} = \sigma \cdot \Delta$ requires σ to scale linearly in V. A value of σ tuned at one V is under-scaled at larger V by the same factor.

5 Experiments

5.1 M4 Hourly, short-term global protocol

Following the M4 leaderboard: per-series z-score normalization, one global model across 414 Hourly series, horizon $H = 48$, best-val checkpoint restored. Model: 2-layer GPT, $d_{\text{model}} = 64$, $h = 4$, block 138, 20k iters, AdamW, temperature-0 argmax at inference.

Soft-CE sweep at $V = 256$.

σ	sMAPE	MASE
0 (plain CE)	18.58	—
1.0	17.55	1.663
2.0	**17.49**	**1.659**
3.0	17.35	1.949

$\sigma = 2$ is the joint-metric optimum. At $\sigma = 3$ sMAPE continues to improve but MASE degrades sharply — a classic over-smoothing signature where forecasts regress toward the conditional mean, rewarded by symmetric sMAPE but penalized by the naive-seasonal-normalized MASE.

Vocabulary sweep (U-curve).

V	σ	sMAPE	MASE
64	0.5	19.96	2.019
128	1.0	19.39	1.888
256	**2.0**	**17.49**	**1.659**
512	4.0	18.21	1.772

Non-monotone with interior optimum at $V = 256$. Smaller V hurts reconstruction precision (bin width doubles); larger V over-refines for the $\sim$ 280k available training tokens. The CE bound is always tight ($\log 2$ per doubling) but the prediction-quality curve is not.

Order-of-derivative ablation.

k	val CE	sMAPE	MASE	note
1	**2.35**	**17.49**	**1.66**	—
2	2.28	39.27	8.93	prediction-error compounding

Token-level training loss is *lower* at $k = 2$, confirming that MASH's higher-order tokenization is learnable. But reconstruction via double integration amplifies model prediction error linearly with the forecast horizon (Prop. 5), collapsing sMAPE. Practically, $k = 1$ is preferred for autoregressive forecasting.

Model-size sensitivity ($V = 256$, $\sigma = 2$).

d_{model}	n_{layer}	params	val CE	sMAPE	MASE
32	2	∼30k	2.36	18.47	1.88
64	**2**	∼130k	2.35	**17.49**	**1.66**
64	4	∼260k	2.35	17.41	1.68
128	2	∼400k	2.38	18.67	1.99

With $\sim$ 280k training tokens, the 64×2 setting ($\sim$ 2 tokens/parameter) is near the data-limited ceiling. Wider (128) overfits (best val at step 2000 of 20000); deeper (64×4) matches within noise at $2.3\times$ training cost.

5.2 ETTh1, long-term multivariate protocol

iTransformer protocol: 7 channels, seq_len = 96, pred_len = 96, chronological 12/4/4-month split, StandardScaler from train. Single model, $d = 128$, $h = 4$, $n_{\text{layer}} = 2$, 2000 iters, batch 32.

config	MSE	MAE
$V = 256$, plain CE	0.9861	0.6056
$V = 64$, plain CE	0.9787	0.6030
$V = 64$, soft-CE $\sigma = 1.0$	0.9806	0.6034

ETTh1 is dominated by a capacity/data-fit floor around MSE = 0.98: vocab and soft-CE move the number by within-noise amounts. Contrast with M4 suggests the *value* of soft targets concentrates in data-starved regimes where many short series force extrapolation of conditional distributions without enough signal to localize a single bin.

5.3 Vocabulary / loss scaling, summary

	per vocab doubling	reference
ideal CE $\mathcal{L}^\star(V)$	$+\log 2 \approx 0.69$ nats	Prop. 7
reconstruction $\|\hat{x} - x\|_\infty$	$\times\frac{1}{2}$	Thm. 1
ordinal smoothing $\sigma^\star$	$\times 2$ (hold value-space constant)	§4

6 Related work

Inverted/channel-mixing transformers. iTransformer [1] and PatchTST [2] remove the autoregressive head and regress the entire horizon in one shot, the dominant design on long-horizon benchmarks. Our model keeps the GPT recipe unchanged; the tokenization carries the inductive bias.

Ordinal and label-smoothed classification. Label smoothing [3] distributes probability uniformly over non-target classes. Distance-weighted variants (cumulative-link ordinal softmax) appear in the ordinal-regression literature; Gaussian soft targets specifically in CORAL [4] and in diffusion-model score matching of discretized targets [5]. Our contribution is situating the smoothing scale σ in a $\Sigma\Delta$-tokenized vocabulary.

$\Sigma\Delta$ quantization. First-order sigma–delta modulators are classical in analog-to-digital conversion [6]; MASH (multi-stage noise shaping) extends to arbitrary order with stability guarantees, originally for audio/oversampling applications [7]. We apply MASH as a differentiable-free preprocessing stage in front of a language model.

7 Future research directions

The experiments here focus on deterministic sMAPE/MSE benchmarks. diff-gpt's architectural choices — autoregressive language-model recipe over a quantized vocabulary — give it structural advantages in a different regime, which we believe deserves dedicated follow-up.

Probabilistic forecasting (quantile / interval prediction). The decoder emits a full distribution over next-token bins at every step; N-sample rollouts give native P10/P50/P90 estimates without a pinball-loss head or quantile-regression machinery. Direct competitors (iTransformer, N-BEATS) optimize MSE, i.e. the conditional mean. Applications: energy-load dispatch with confidence bands, financial risk (VaR, expected shortfall), reliability engineering, weather ensembles, inventory planning.

Heavy-tailed and multi-modal conditional distributions. ℓ_2 regression collapses toward the conditional mean and fails catastrophically on bursty / spike / regime-change series where the true $p(X_t \mid \text{ctx})$ is bimodal ("jump up or jump down, not in between"). A categorical head models this directly. Candidate domains: network traffic, high-frequency finance (CDS spreads, order-book imbalances), industrial sensors with fault modes.

Foundation-style zero-/few-shot forecasting. The LM recipe admits trivial pretrain-then-transfer: one model pretrained on a million heterogeneous series, few-shot on a new one. This is the angle pursued by Chronos, TimesFM, MOIRAI. Our tokenization contract (two closed-form scaling laws) is theoretically cleaner than their binning schemes, which is a plausible advantage at scale.

Heterogeneous / multi-resolution series in one model. The encoder self-normalizes per channel via D, letting a single parameter set handle μV ECG, m/s^2 seismograms, and dollar-scale macro in one training run. Regression baselines need per-channel heads or manual unification. For foundation models this is a core problem and our scheme is structurally simpler.

Anomaly detection via conditional likelihood. $-\log p(\text{token}_t \mid \text{ctx})$ is available for free at each step and calibrates correctly under variable conditional variance. Regression-based approaches rely on $\|y_t - \hat{y}_t\|$, which miscalibrates on heteroscedastic series. Candidate domains: industrial monitoring, network intrusion, patient monitoring.

Conditional / counterfactual generation. "What if X_t jumps by Δ next week?" reduces to token substitution in the prefix, then resume generation. Regression models need a separate conditional mode or structured covariate mechanism.

Edge inference. A 130k-parameter model attains the results in Section 5; the bulk of the inductive bias lives in the tokenization, not the weights. This makes sub-mobile-CPU inference realistic for production forecasting on IoT or embedded devices — a regime where Transformer SOTA typically does not fit.

Where the architecture will not win. Deterministic MSE benchmarks at long horizons (ETTh1 / Traffic at pred_len ≥ 336) are structurally unfavorable: one-shot inverted transformers emit the whole horizon without autoregressive error compounding, and Prop. 5 guarantees our error grows at rate H^{k-1}. Deterministic long-horizon forecasting is not a target; the wins above are.

8 Conclusion

A decoder-only transformer trained autoregressively on derivative-plus- MASH-quantized tokens gives competitive short- and long-term forecasts without a specialized head. The tokenization contract provides two closed-form vocabulary-scaling laws valid at any derivative order; prediction-error compounding gives a principled preference for $k = 1$ in forecasting despite MASH's tight higher-order reconstruction bound. Ordinal soft cross-entropy matches the natural bin-wise geometry and yields a 6% relative sMAPE reduction on M4 Hourly without architectural change.

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